

Trinity Demonstrator Note 1: Event Cleaning

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August 26, 2025

1 Script Location and Instructions

The link to the [GitHub](#)

The event cleaning script is located here on hive:

```
/storage/hive/project/phy-otte/shared/Trinity/DataAnalysis/EventCleaning/  
ClusterCleaning/ClusterCleaning  
./ClusterCleaning YYYYMMDD
```

The script can be edited here:

```
/storage/hive/project/phy-otte/shared/Trinity/DataAnalysis/EventCleaning/  
ClusterCleaning/ClusterCleaning/EventCleaning_Clusters.cpp
```

```
/storage/hive/project/phy-otte/shared/Trinity/DataAnalysis/EventCleaning/  
ClusterCleaning/ClusterCleaning/EventCleaning_Clusters.h
```

```
make
```

The output files are created for pdfs and .root files located here

```
/storage/hive/project/phy-otte/shared/Trinity/DataAnalysis/EventCleaning/  
ClusterCleaning/Output
```

The script is used in the data summaries and there are scripts that can run all the event cleanings. This script will rerun every event cleaning based on the dates you chose (hardcoded) and will send jobs to Hive to fulfill them.

```
/storage/hive/project/phy-otte/shared/Trinity/DataAnalysis/  
EventCleaning/allEventcleaning.py
```

Then create a large single pdf with

```
/storage/hive/project/phy-otte/sstepanoff3/EventCleaning/CombineCleaning  
./CombineCleaning
```

2 How to interpret the data

2.1 Camera Mapping

The camera mapping describes the physical orientation of camera electronics and how to address them for data analysis. Figure 1 shows the camera mapping for locating pixels (SiPMs), MUSICs, SIAB, HV channels and more.

Let's start with key terms:

- **SiPM:** This is the chip that holds 16 pixels silicon multipliers, which register the photons that hit the surface. These are not labeled directly on the camera mapping but refer to the 16 pixels contained in one SIAB.

Each Event - Signal

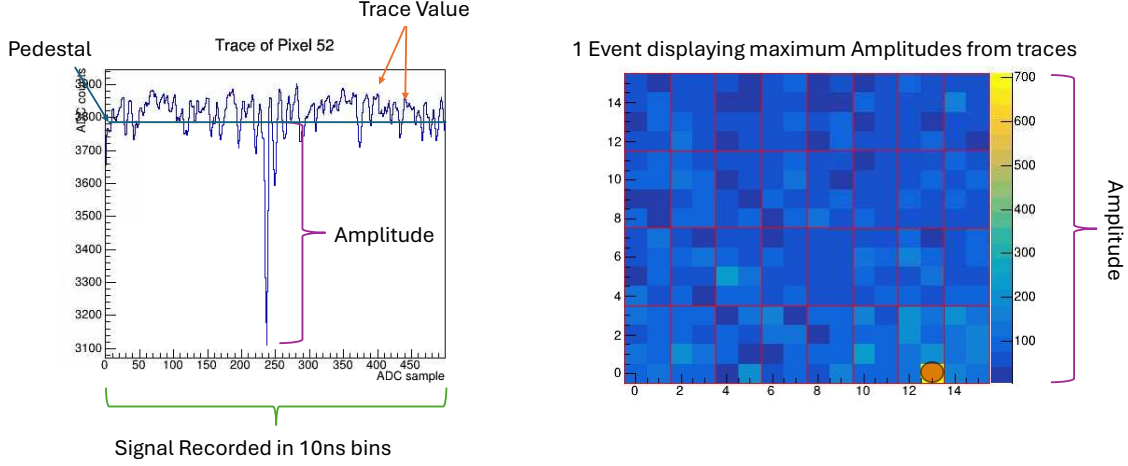


Figure 2: Left: Trace of a signal. Trace value shown with orange arrows it is the blue jagged line. The pedestal is an average of the signal noise. Amplitude is the signal difference from the pedestal. ADC counts are the signal and ADC sample are time bins of 10 ns. Right: Image of all 256 pixels as they are laid out in the physical camera. The colors refer to the max amplitude in ADC in that pixel with blue being less and yellow being more. The orange circle shows the pixel that is shown on the left.

2.2 Root Data

We collect data in ROOT files. This file type allows for multiple subfolders of data to be stored within one file, enabling efficient access to large amounts of data. For more information about the ROOT structure, see [this website](#). A single file for the Trinity Demonstrator contains approximately 97 seconds of data acquisition. The data is saved as 5.12 s events, and the number of events per ROOT file depends on how often the data acquisition triggers. Each event records the signal trace for each SiPM, as shown in Figure 2 Left. This pixel is an example of a pixel that successfully registered a signal and triggered the camera readout. Each time bin represents 10 ns. The signal of interest appears as a dip around 236–242 time bins. The noise before and after the dip is referred to as the pedestal. The amplitude is calculated as the difference between the peak and the pedestal:

$$\text{Amplitude} = \text{Trace}_{\text{PeakTimeBin}} - \text{Pedestal}$$

The pedestal is computed using the following formula, using the first 100 time bins as the sample:

$$\text{Pedestal} = \frac{\sum_{i=0}^{100} \text{Trace}_i}{100}$$

The first 100 times bins are well before the signal is expected so we are never sampling signal in the pedestal. It is also useful to calculate the pedestal RMS to account for noise variation. On the right side of Figure 2 is a full camera display of an event, which shows the maximum amplitude values of all pixels.

In Figure 3, an example of a trace from a pixel containing only noise is shown. In this case, there is no clear dip around the 235–245 ns sample bins. Instead, the amplitude barely falls below the noise level. The trace from pixel 49 would have similar structure as the traces from the other 254 pixels that do not show a signal.

Signal amplitude is reported as Analog-Digital Converter counts. The analog signal is the electrical signal produced in the SiPMs by amplifying the photoelectron signal produced from photons incident with the SiPM. This is converted to a digital signal in the MUSICs. The conversion between ADC count and photoelectrons produced in the SiPM is 24.1:1.

To extract these data from the data ROOT files, we use the C++ library [ExACT](#).

Each Event - Noise

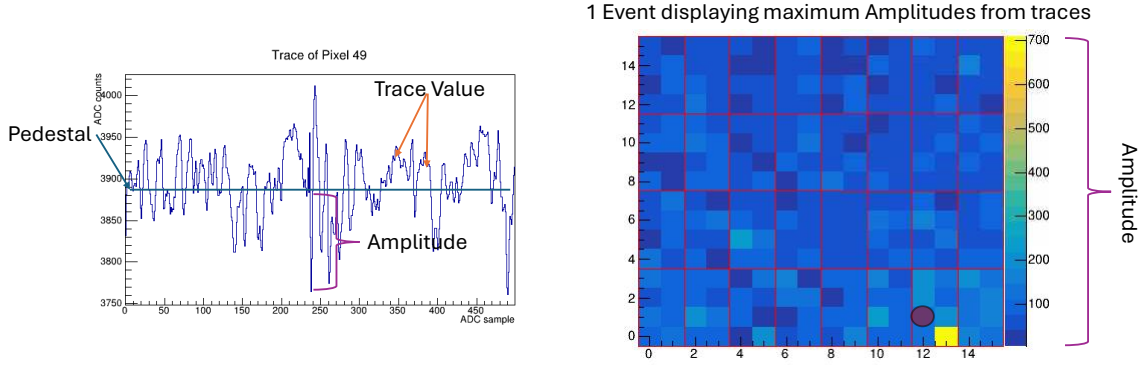


Figure 3: Left: Trace of a pixel with noise only. Right: Image of all 256 pixels as they are laid out in the physical camera. The purple circle shows the pixel that is shown on the right.

2.3 Crosstalk

Crosstalk is a hardware issue where the signal from one SiPM pixel is electronically echoed to other pixels connected to the triggered one. The crosstalk will present as delayed peaks in the trace. This crosstalk in SiPMs is described in [?]. We use the same SiPMs as SPB2. All events with a triggered pixel will have crosstalk. However, we need to identify what crosstalk and the real signal are. Crosstalk will have a lower amplitude than the triggered pixel trace and will occur in a later time bin. When two pixels have an amplitude that peaks at the same time bin as the triggered pixel time bin, it is not caused by crosstalk. Here is an example of some crosstalk. In figure 4, the second SIAB from the top left shows up on the event display as a lighter blue, indicating a higher amplitude overall, and the yellow pixel is the triggered pixel. A triggered pixel to raise the amplitude of all the pixels in its MUSIC and SIAB is a typical response of our system. Looking at the trace of three different pixels, the triggered pixel (Figure 5), a pixel inside the SIAB (Figure 6), and a pixel outside the SIAB (Figure 7) we can understand what the crosstalk pixel will show us. When looking at the triggered pixel (Figure 5) and pixel inside the SIAB (Figure 6), the dots that indicate the peak time bin (green), initiation of signal (purple) and peak (blue). The trace of the triggered pixel will always have the peak time bin and initiation of the signal before the crosstalk peak time bin and onset. Crosstalk can not happen until after the pixel has been triggered. As seen in blue, the amplitude will be significantly lower as photons do not generate the signal. When looking at the noise pixel 7, the amplitude is never registered below the pedestal, and there is no crosstalk amplitude signal. Allowing for a distinct difference between crosstalk pixels and the noise pixels.

How does this affect the event cleaning? Events with pixels surviving in only one SIAB could only be from crosstalk, making them happen to survive instead of being a real airshower event. A case of a surviving crosstalk event can be seen in this sample cleaned event in figure 8. The amplitude in PEs for the yellow is 10 PE, which means all those yellow pixels are relatively low in amplitude, and only the yellow pixels are contained in the same SIAB. We would not consider this an event worth looking into further. If you need further proof, you can always look at the traces.

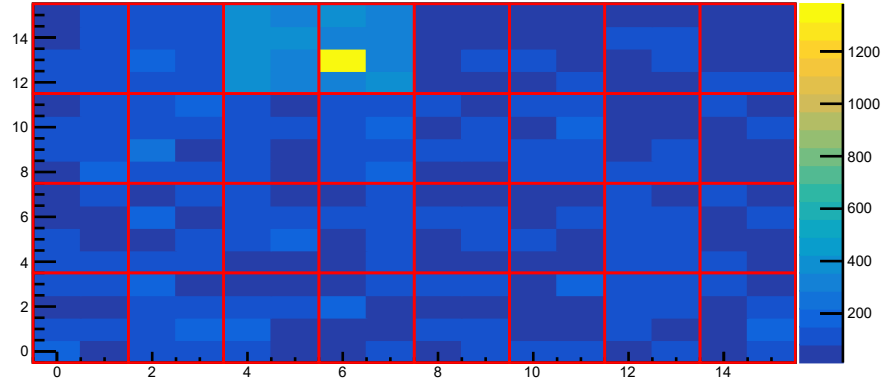


Figure 4: Camera Event Display that shows the second SIAB from the top left as having more amplitude than other SIABs this is from cross talk of the triggered yellow pixel. Amplitudes are in ADC counts.

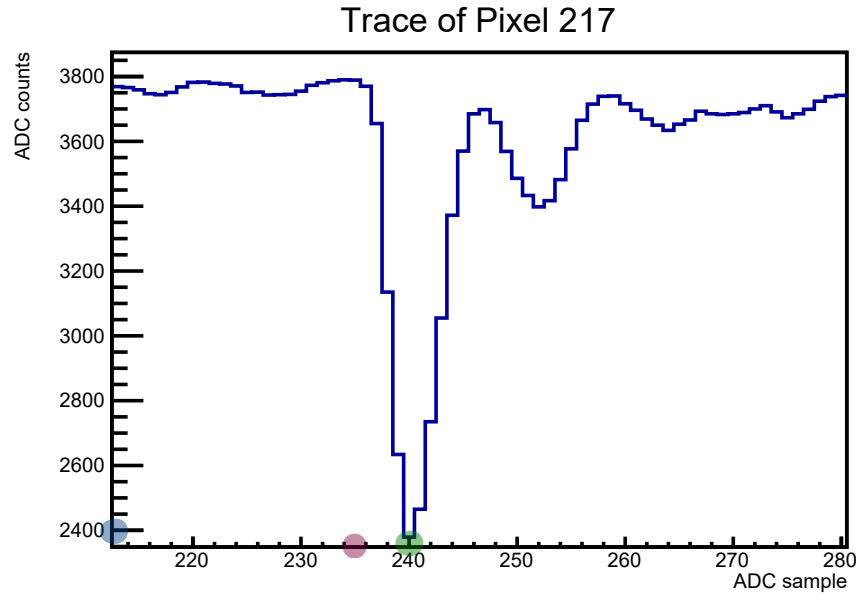


Figure 5: Trace of the triggered pixel from figure 4. There are three dots green is for where the signal peaks, purple is where the initiation of the signal starts and blue shows the peak of the signal in ADC. This trace is showing just a trace of a triggered pixel. Note that the samples are only showing the area of interest.

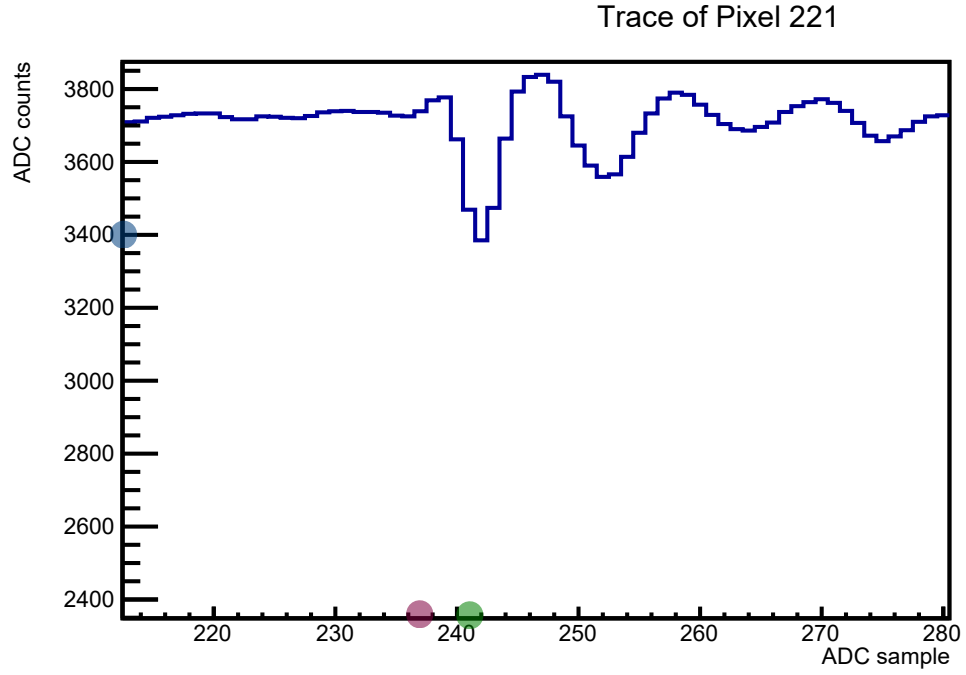


Figure 6: Trace of the pixel to the right of the the triggered pixel from 4. There are three dots: green is for where the signal peaks, purple is where the initiation of the signal starts and blue shows the peak of the signal in ADC. This trace is showing a cross talk trace from the triggered pixel. Note that the samples are only showing the area of interest.

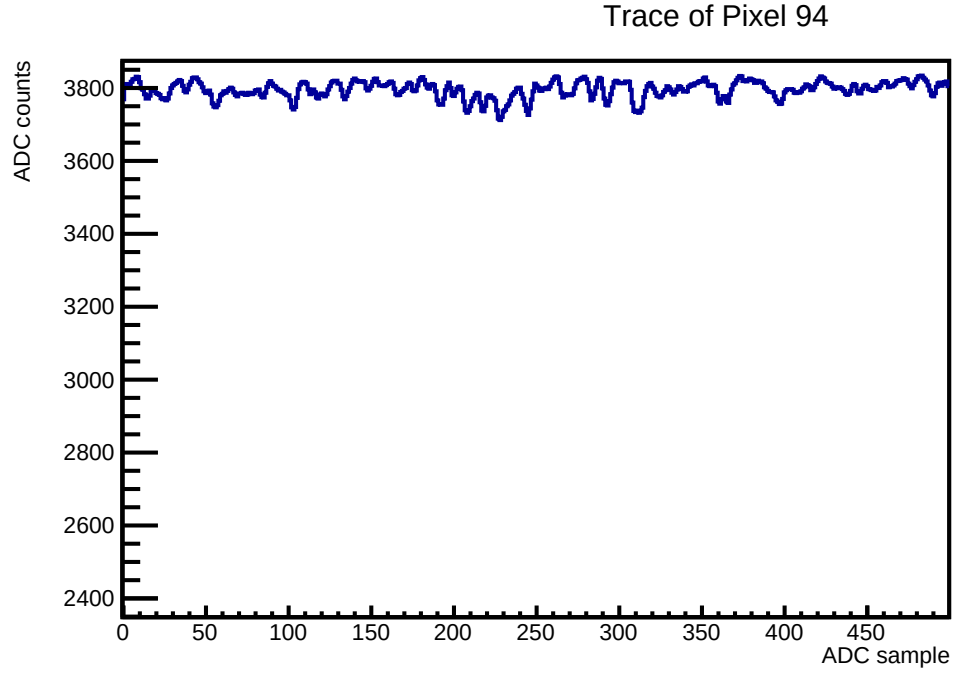


Figure 7: Trace of the noise pixel from 4 that is not in the same SIAB as the cross talk. This is showing the trace of a noise pixel.

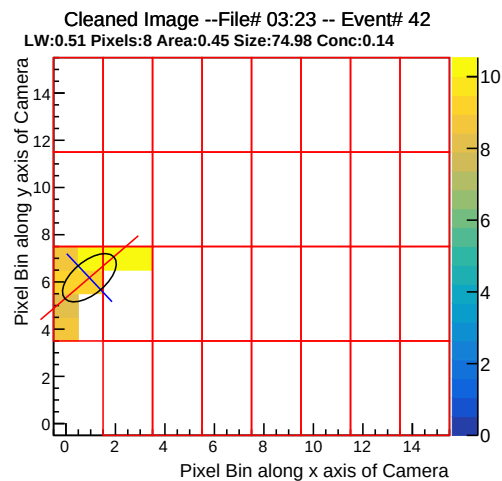
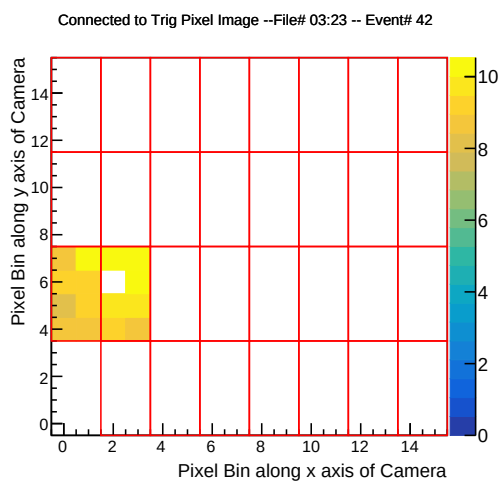
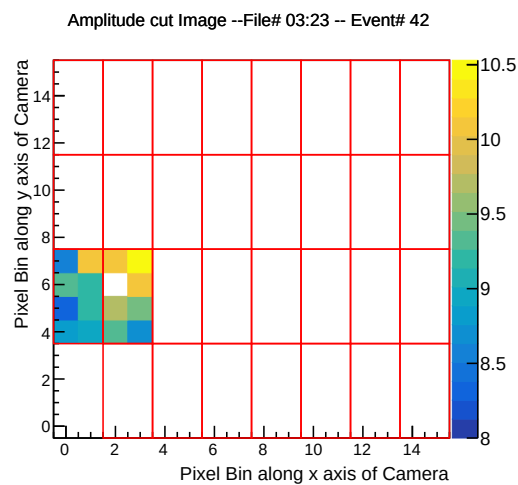
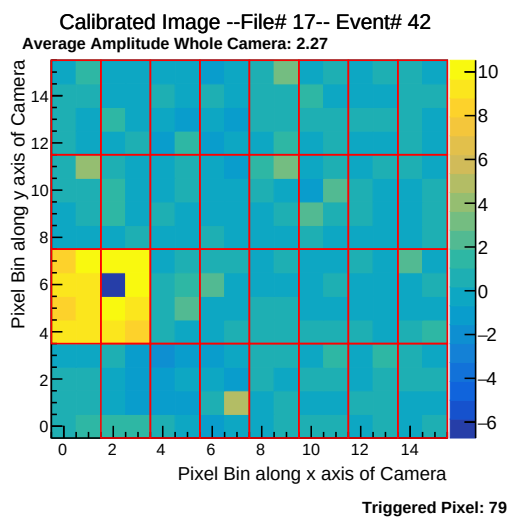


Figure 8: Event Cleaning display showing a cross talk event surviving the event cleaning.

3 How the Event Cleaning Works

The event cleaning process involves executing a series of cuts on data taken throughout the night of observations to eliminate known background events, leaving only those that may originate from non-background sources. Common types of background include muons, background fluctuations in the night sky, and crosstalk. Event cleaning consists of two main sections: the actual events and useful histograms. It is anticipated that between 0 to 4 events will survive after a night of data collection. The event cleaning product per event looks as seen in Figure 4 for the four-panel plot. The first panel displays the calibrated original event image, representing the signal's amplitude above the noise, with colors indicating photoelectrons. Each square in this image corresponds to a pixel of the camera. The second panel presents a straightforward amplitude cut, showing all pixels at a specific time bin of the trace that exceeds a designated threshold amplitude. The third panel, labeled "Connected to Trigger Pixel Image," depicts the amplitude of a pixel and whether it is connected via borders to the triggered pixel. Finally, the cleaned image is the last plot, displaying only the pixels that have survived all previous cuts. More on each of these panels will be described below. Any time a cut condition is not met the event is no longer processed to save computational time. Thus, if the event does not survive to the fourth plot, no plot will be produced.

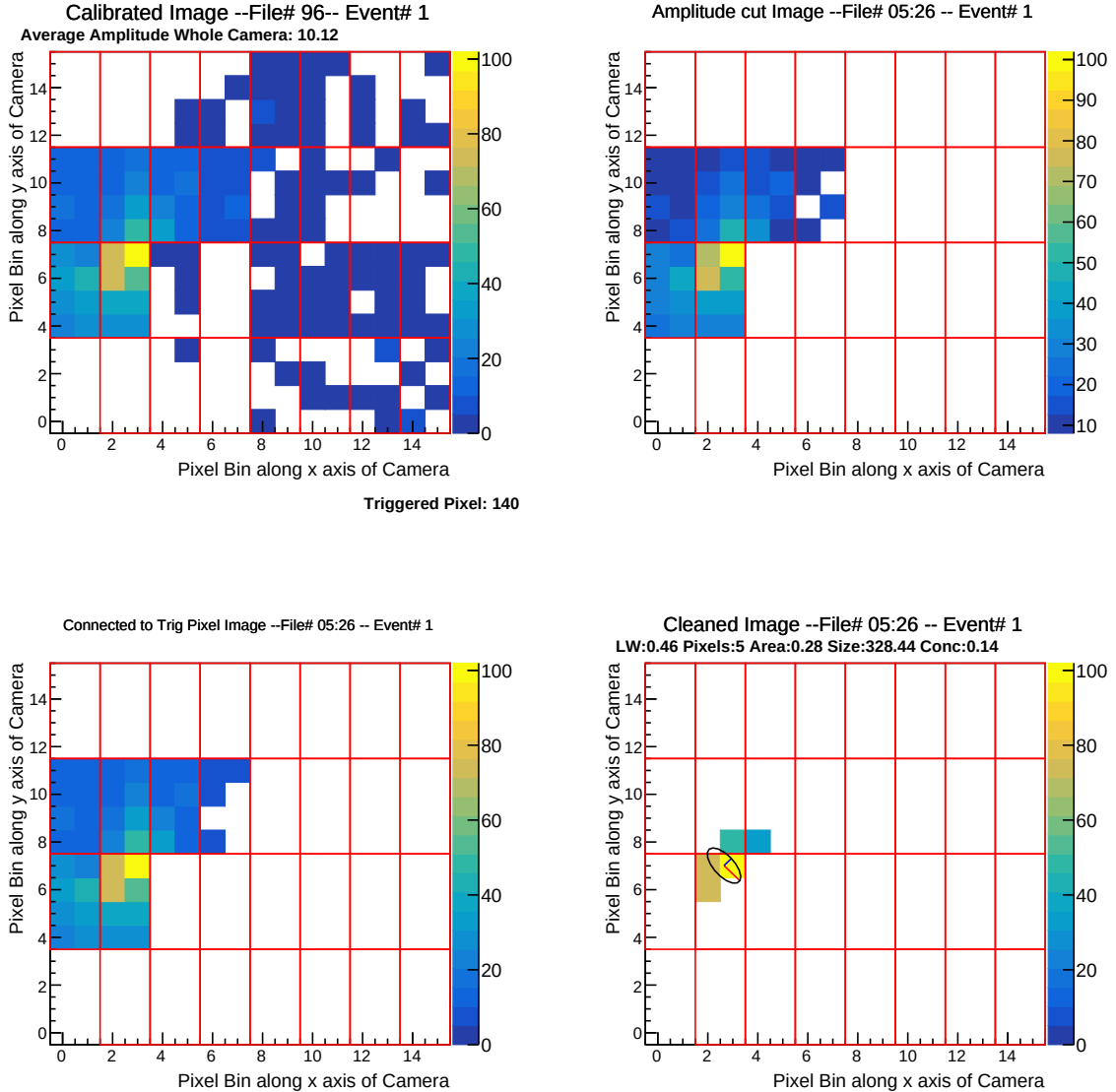


Figure 9: Typical 4 panel plot that shows the event cleaning process for the data summaries Each figure will be explain in detail below. Event is from 2025-02-26

3.1 Gain Calibrated Image (Top Left)

1. **Applying Gain Calibration from Flasher:** The event cleaning process begins with the gain calibration of the event using flasher data, as illustrated in figure 10. While the gain calibration is not a pre-defined cut, the camera must be calibrated to accommodate slight variations in pixel response, thereby improving event cleaning accuracy. Flasher data is generated when a uniform flash of light lights the camera at a frequency of 1 Hz. The flasher lets us simultaneously understand the camera's response to a consistent known illumination flash across all pixels throughout data collection. Consequently, on average, pixels that exhibit higher amplitude responses to the flasher light (indicated in yellow) will have their values slightly reduced in the gain calibrated image, while pixels displaying lower amplitude responses (marked in blue) will receive a slight boost to their peak amplitudes. The gain calibration factor is calculated using the formula:

$$Gain_{rel_p} = \frac{\sum_{All\ events} AvgAmp_p}{Median\left(\sum_{All\ events} \sum_{All\ pixels} AvgAmp_p\right)}, \quad p = \text{pixel}.$$

This gain calibration factor is subsequently applied to the data collection event, resulting in the creation of the gain calibrated image (GCI) through the equation:

$$GCI_p = \frac{Amp_p}{Gain_{rel_p}}.$$

The final outcome of this calibration process can be observed in Figure 11. Notably, in the top right SIAB, which predominantly shows blue, the calibrated event reflects an amplified signal as a result of this calibration.

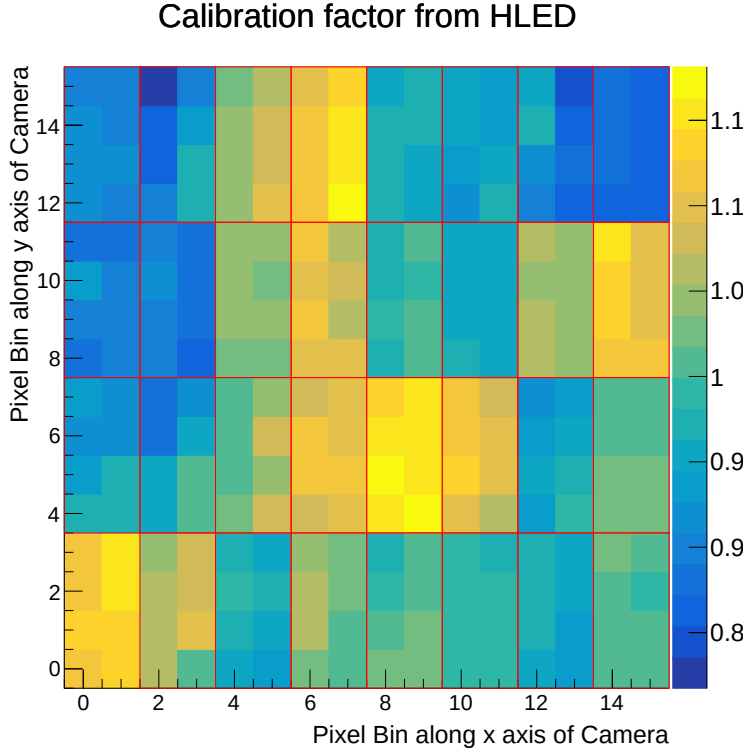


Figure 10: Calibration factors from the night of 2025-02-24 (HLED means health LED aka the flasher). The colorbar is the calibration factor that will be applied to the pixel. Yellow means that the amplitude from the pixel has a higher amplitude than average and blue means that the amplitude from the pixel has a lower amplitude.

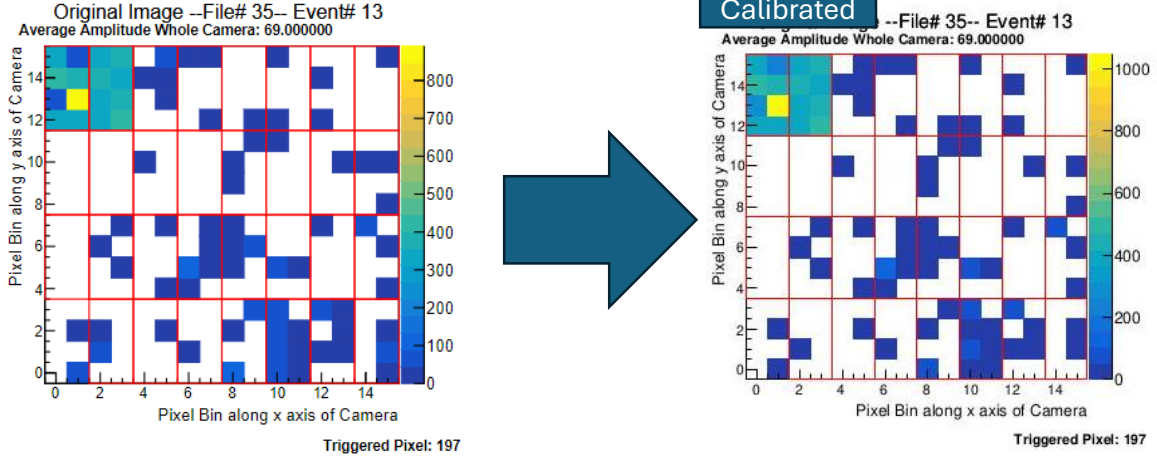


Figure 11: Left: Shows the whole camera event before the calibration factor applied. Right: Shows the whole camera event after the calibration so that pixels that were blue in the calibration mean will get a signal boost as seen in the color bar and top left SIAB when used with figure 10.

2. **Flasher Event vs. Data Events:** When the system receives a trigger, it records the event, and the event builder can save the pixel traces in four branches of the root files: Test, HLED, Forced, or Bifocal. For our purposes, we only care about the Test and HLED branches. To simplify, data from the Test branch is used during normal data collection to record events, while the HLED branch is designated for recording flasher events.

However, the event builder can make mistakes when classifying whether an event belongs in the Test or HLED branch. The event cleaning process categorizes every event into Flasher events and Data events. This classification is determined by analyzing the average max pixel amplitude of the entire event and creating a distribution for that night, as illustrated in Figure 12. The average amplitude of the camera due to the flasher events is expected to be greater than 15 photoelectrons (PEs). Any event with an average amplitude below 15 PEs is expected to be a Data event and is put through the data cleaning. Flasher events show higher averages since all pixels have elevated amplitudes, resulting in a higher overall event average.

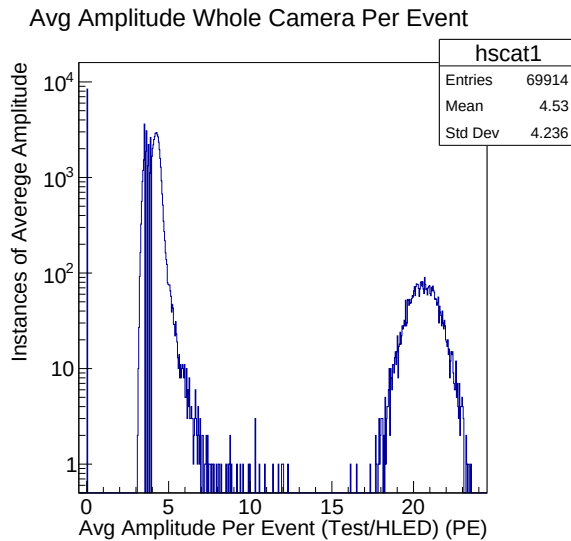


Figure 12: Distribution of Average Amplitude of all pixels per event over a night. Two distinct average amplitudes have two distributions the one centered around 5 PE are regular data events and the one centered around 21 is from Flasher events.

3. **Triggered pixel has sufficient amplitude:** The triggered pixel, identified by locating the highest amplitude pixel within the Triggered SIAB, must exhibit an amplitude greater than 8 Photoelectrons (PE) at time bin 239. This threshold of 8 PE is determined as this is where we

can comfortably discern events from noise to a signal. The specific time bin is established by identifying the moment when most trace peaks over events spanning an entire night, as illustrated in Figure 13. We focused on determining the most frequently occurring peak time bin within the data and selected the earliest instance of this peak. This systematic approach ensures we accurately pinpoint the region where the amplitude will show the rise of the maximum amplitude and eliminate some crosstalk events.

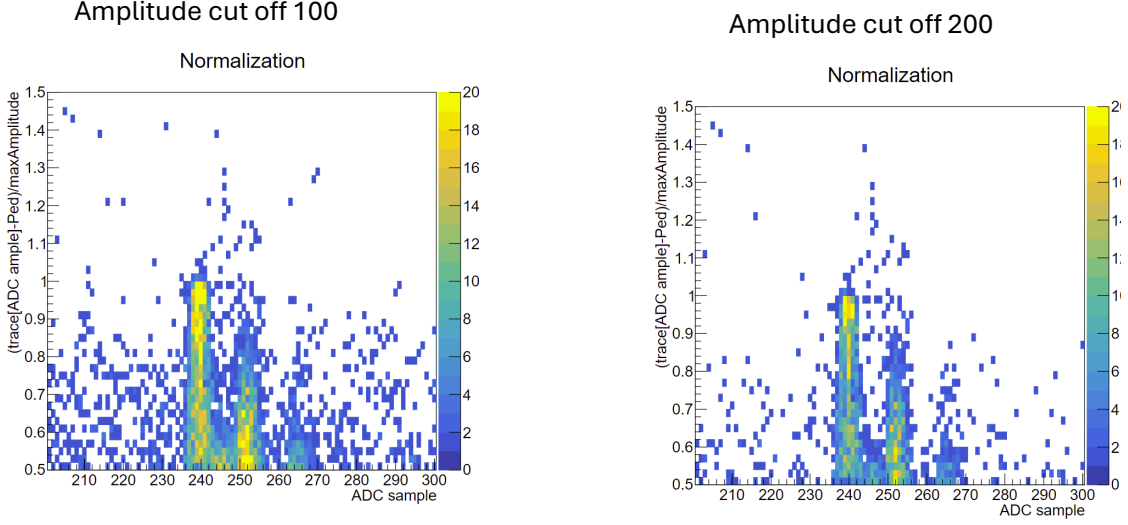


Figure 13: Left: For each trace the amplitude at that time bin is plot and averaged over the max amplitude. Only values where the amplitude was above 100 ADC show in this plot the whole trace is saved in this histogram. A large peak of events that are at the max amplitude happen at 239-241 time bins. This represents max ampiltudes due to signal not noise. Everything else is noise or crosstalk. Right: same as the left except the cut off is 200 ADC counts.

SUMMARY TABLE FOR CALIBRATED IMAGE (TOP LEFT)

Item	Cut/Apply	Notes
Flasher Calibration	Apply	Amp > 8 PE
Separate Events	Apply	
Triggered Pixel	Cut	
High enough HV currents	Cut	

3.2 Amplitude Cut Image (Top Right)

1. **Non Triggered Pixel cut off:** Every pixel with an amplitude above 8 PE at time bin 239 regardless of position is saved and plotted in the Amplitude cut image.
2. **Three pixels survived:** After the pixel cut at least 3 need to survive to continue the event cleaning.

SUMMARY TABLE FOR AMPLITUDE CUT IMAGE (TOP RIGHT)

Item	Cut/Apply	Notes
All Pixel Amplitude	Cut	Amp > 8 PE
Three Surviving pixels	Apply	

3.3 Connected to Trig Pixel Image (Bottom Left)

The third cleaned event camera plot shows the surviving pixels connected to the triggered pixel, all share the earliest time bin for their amplitudes.

1. **Triggered Pixel Connection** We will only retain pixels connected to the triggered pixel through broadside connections, as illustrated in Figure 14. The triggered pixel is represented in yellow, the broadside pixels are shown in orange, and the grey pixels indicate other nearby pixels. For an air shower event, we expect the pixels to be connected since the light will continuously shine across the sky. Discrete groups of pixels not connected to the triggered pixel are expected to be noise.

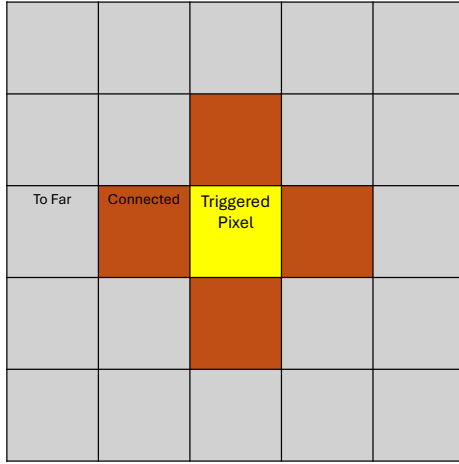


Figure 14: Illustration of pixels on the camera to demonstrate broad side connection to the triggered pixel. Triggered pixel in yellow, and broadside connecting pixels in orange and gray are pixels that are too far away, to be saved.

2. **Pixels have the same amplitude time bin:** Once we identify all the pixels connected to the triggered pixel, we aim to reduce the number of surviving crosstalk events. One way to identify crosstalk is by observing delayed signals. In Figure 15, it can be seen that the crosstalk event increases in amplitude after the initial signal from the triggered pixel begins. By focusing on signals with the same earliest peak amplitude time bin, we can eliminate some crosstalk events. We have a tolerance of the same time bin as we do not expect the event to last beyond 10 ns.

Example: For the connected pixels, we have the following data: pixels = [11, 14, 15, 19, 77, 80, 81] and their respective time bins = [238, 240, 238, 239, 238, 238, 240]. This means that the pixels 11, 15, 77, and 80 will move on to the next stage of the cleaning process.

3. **Three pixels survived:** After the pixel cut at least 3 need to survive to continue the event cleaning

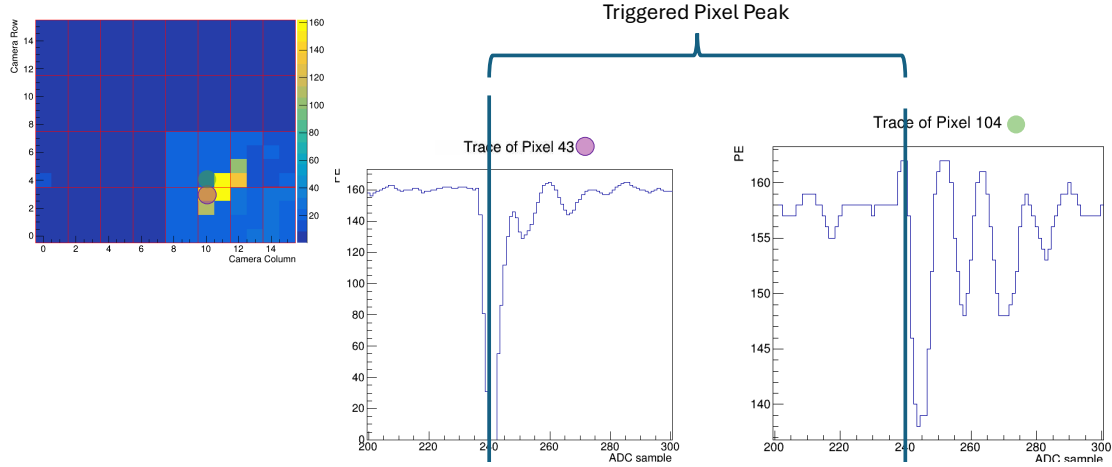


Figure 15: From one event in the top left, looking at cross talk between the triggered pixel (middle trace) and the pixel above (right trace) showing where the peak time bin is for the triggered pixel and the crosstalk.

Summary Table for Connected to Trig Pixel Image (Bottom Left)

Item	Cut/Apply	Notes
Triggered Pixel Connection	Cut	Figure 14
Same Peak Time Bin	Cut	Follow Earliest Surviving Time Bin
Three Surviving pixels	Apply	

3.4 Cleaned Image (Bottom Right)

1. **PCA:** PCA details will be in section 3.4.2. There is some randomization of the location of the pixels in our PCA. The slight randomization is because actual photons can hit anywhere on a pixel and will still read out that pixel. So the x and y camera positions have a random value $[-0.49, 0.49]$ added to them displayed in figure 16 left. With the randomized locations the length/width (LW) ratio that is found from the PCA is slightly varied following a more realistic scenario seen in figure 16 right (LW is discussed more below in 3.4.1). From the PCA LW ratio analysis, we can cut too circular data. Since crosstalk events are expected to create a more circular ellipse, we can remove events that have LW ratios greater than 0.5. The LW ratio is the final cut to the data. An event that survives this will create the four-panel plot the user sees.

3.4.1 Terms on the Final Plot

1. **LW (Length Width Ratio):** is when you take the ratio of the length (PCA moment 1) and width (PCA moment 2) and get the difference.

$$LW = \sqrt{\lambda_2}/\sqrt{\lambda_1}$$

This can describe how elliptical an event is. 0 being a line and 1 being a circle.

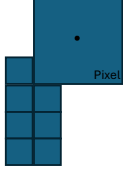
2. **Pixels:** The number of pixels that survived the cleaning.
3. **Size:** Sum of the surviving pixel amplitude

$$size = \sum_p AmpSurviving_p$$

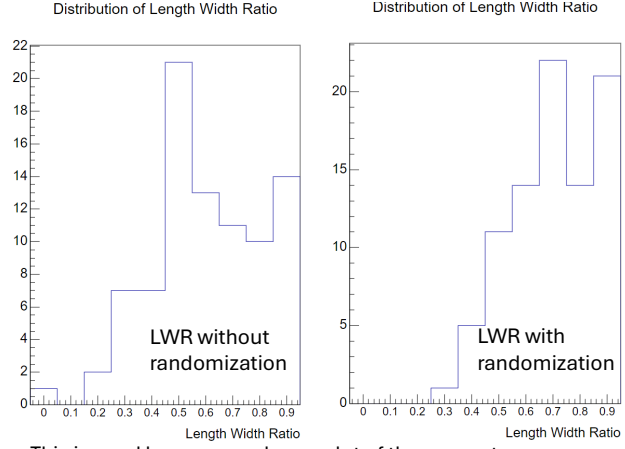
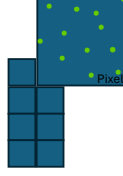
can hit anywhere on the pixel

```
for (int w = 0; w < PEs ; w++){
    std::vector<double> randomNumbers = generateRandomNumbers();
    double data[2] = {nx + randomNumbers[0], ny+randomNumbers[1]};
    pca->AddRow(data);
}
```

Prior model all PE map to center of pixel



New model each PE gets a different location within the pixel



This is good because we know a lot of these events are Cross talk so the shift towards more circular

Figure 16: Left: Showing the code that loads the data into the PCA class so that each x and y position is slightly randomized to be similar to real life photon hits. Then below shows how the randomization changes the x and y position from the pixel center so there is the randomization include. Right: Distributions of the LW ratio before and after randomization. Events got more round which is to be expected.

4. Conc(Concentration):

$$conc = \frac{Amp_{Triggerp}}{size}$$

5. COG: While not visible it is where the PCA ellipse is centered

6. **M3Long:** A factor that describes the asymmetry of the event allow the us to understand which way the shower direction is oriented or head and tail of a shower. Its the difference between the peak of the amplitude in the (x,y plane) and COG. The sign will tell you which way the shower is heading. The shower will start with higher amplitudes and head away from there.

$$M3Long = \left(\frac{1}{N} \sum_{i=i}^N q_i^3 (x_i - x_{COG})^3 \right)^{1/3}$$

where i is the pixel index, N is the number of surviving pixels, q_i is the amplitude contained in pixel i, x_{COG} is the COG of the event. This for both the primary and secondary moments so the x and y and it is displayed to the user as (x.xx, x.xx). So when the values are (+,+) its going up and to the right in the (x,y) plane and when its (+,-) its going down and to the right in the (x,y) plane.

Summary Table for Connected to Trig Pixel Image (Bottom Left)

Item	Cut/Apply	Notes
LW Ratio	Cut	LW < 0.5

Table 1: An example table.

3.4.2 PCA

Principle Component Analysis is a method to fit data that will determine variance along each feature (for us this is x,y coordinates). The matrix of data is created by taking the pixel location and added it to the matrix as many times as the amplitude making it weighted.

```
for pixel:
    for weight:
        data append {xpos, ypos}
```

To perform PCA you need to get the covariance, this is sourced from the source code of root for the covariance matrix. Root method for PCA forces normalization which does not allow the eigen values to be correct. From there root has built in functions to calculate the eigenvalues, eigenvectors and sigmas. The Eigenvector provide angle of the first moment and the second moment will always be 90° by PCA theory and eigenvectors are orthogonal. Then length and width of the moments is found with the eigenvalues from the $Length = \sqrt{\lambda_1}$ seen as the red line and $Width = \sqrt{\lambda_2}$ seen as the blue line as seen in figure 17 and the shower axis is $ShowerAxis_{major} = 2Length$ and $ShowerAxis_{minor} = 2Width$.

The shower axis can be seen at the vector lines that extend beyond the ellipse. The ellipse is created by using the length and width as the semi-major and semi-minor axis of the ellipse and then the angle is retrieved from the first eigenvector.

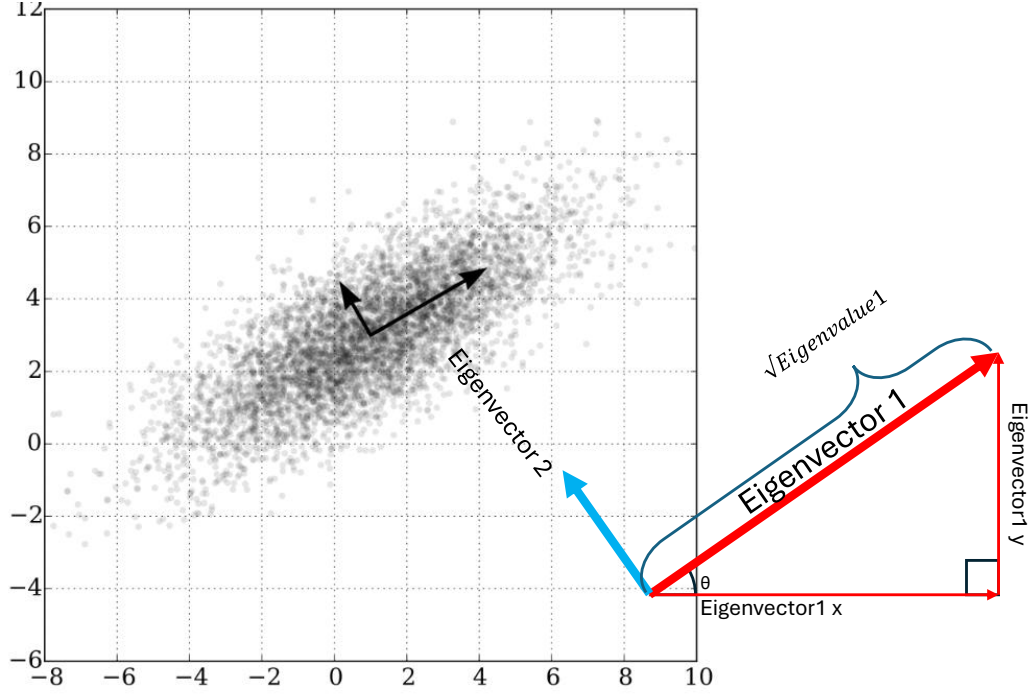


Figure 17: Sample data that from PCA wiki page that shows the PCA arrows on the broken down.